

20/3/24
Wednesday

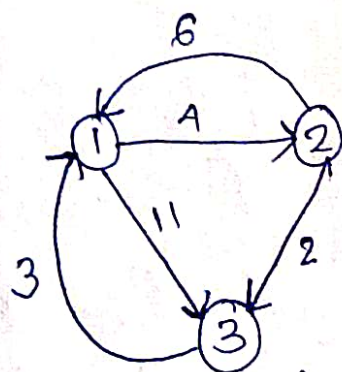
Module - 3

→ Dynamic programming

→

All pair shortest path problem

Q. Find the minimum cost for the following graph using all pair shortest path problem.



$$A_0 = \{0\}$$

$$A_1^{(via)} = \{0, 1\}$$

$$\min(\infty, *) = 7 \checkmark$$

$$A_2 = \{0, 1, 2\}$$

(via 2)

	1	2	3
1	0	4	11
2	6	0	2
3	3	∞	0

	1	2	3
1	0	4	11
2	6	0	2
3	3	7	0

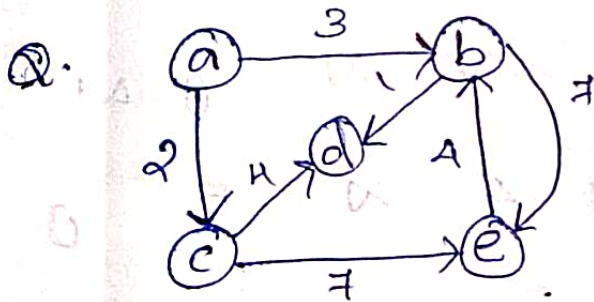
	1	2	3
1	0	4	6
2	6	0	2
3	3	7	0

$$A_3 = \{0, 1, 2, 3\}$$

(via 3)

	1	2	3
1	0	4	6
2	5	0	2
3	3	7	0

Find the cost of $\{A, E\}$



$$A_0 = \{0\}$$

	a	b	c	d	e
a	0	3	2	∞	∞
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	∞	0

$$A_1 = \{0, a\}$$

	a	b	c	d	e
a	0	3	2	∞	∞
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	∞	0

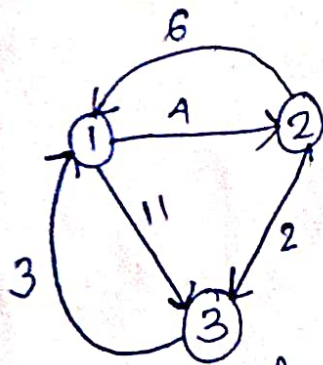
20/3/24
Wednesday

Module - 3

→ Dynamic programming

→ All pair shortest path problem

Q. Find the minimum cost for the following graph using all pair shortest path problem.



$$A_0 = \{0\}$$

$$A_1^{(via)} = \{0, 1\}$$

$$\min(\infty, 7) = 7 \checkmark$$

$$A_2 = \{0, 1, 2\}$$

(via 2)

	1	2	3
1	0	4	11
2	6	0	2
3	3	∞	0

	1	2	3
1	0	4	11
2	6	0	2
3	3	7	0

	1	2	3
1	0	4	6
2	6	0	2
3	3	7	0

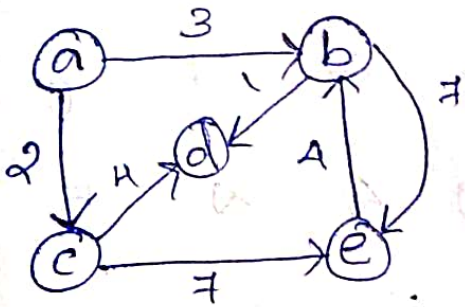
$$A_3 = \{0, 1, 2, 3\}$$

(via 3)

	1	2	3
1	0	4	6
2	5	0	2
3	3	7	0

Find the cost of $\{A, E\}$

Q.



$$A_0 = \{0\}$$

	a	b	c	d	e
a	0	3	2	∞	∞
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	∞	0

$$A_1 = \{0, a\}$$

	a	b	c	d	e
a	0	3	2	∞	∞
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	∞	0

$$A_2 = S_1$$

$$A_2 = \{0, a, b\}$$

	a	b	c	d	e
a	0	3	2	4	10
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	5	0

$$A_3 = \{0, a, b, c\}$$

	a	b	c	d	e
a	0	3	2	4	9
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	5	0

$$A_4 = \{0, a, b, c, d\}$$

	a	b	c	d	e
a	0	3	2	4	9
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	5	0

$$A_5 = \{0, a, b, c, d, e\}$$

	a	b	c	d	e
a	0	3	2	4	9
b	∞	0	∞	1	7
c	∞	∞	0	4	7
d	∞	∞	∞	0	∞
e	∞	4	∞	5	0

Q. a to e?

Ans: 9

path: a, c, e

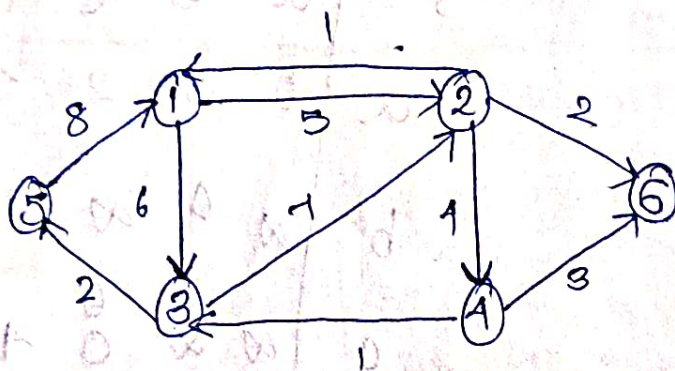
$$\text{cost} = 2 + 7 = \underline{\underline{9}}$$

21/3/24
Thursday

Q. Find the minimum cost using all pair shortest path problem for the following graph

i) 1 - 6

ii) 4 - 1



$$A_0 = \{0\}$$

	1	2	3	4	5	6
1	0	5	6	∞	∞	∞
2	1	0	∞	4	∞	2
3	∞	7	0	∞	2	∞
4	∞	∞	1	0	∞	3
5	8	∞	∞	∞	0	∞
6	∞	∞	∞	∞	∞	0

$$A_1 = \{0, 1\}$$

	1	2	3	4	5	6
1	0	5	6	∞	∞	∞
2	1	0	7	4	∞	2
3	∞	7	0	∞	2	∞
4	∞	∞	1	0	∞	3
5	8	13	14	∞	0	∞
6	∞	∞	∞	∞	∞	0

9
15

$$A_2 = \{0, 1, 2\}$$

	1	2	3	4	5	6
1	0	5	6	9	∞	7
2	1	0	7	4	∞	2
3	8	7	0	11	2	9
4	∞	∞	1	0	∞	∞
5	8	13	14	17	0	15
6	∞	∞	∞	∞	∞	0

$$A_3 = \{0, 1, 2, 3\}$$

	1	2	3	4	5	6
1	0	5	6	9	8	7
2	1	0	7	4	9	2
3	8	7	0	11	2	9
4	9	8	1	0	3	10
5	8	13	14	17	0	15
6	∞	∞	∞	∞	∞	0

$$A_4 = \{0, 1, 2, 3, 4\}$$

	1	2	3	4	5	6
1	0	5	6	9	8	7
2	1	0	5	4	7	2
3	8	7	0	11	2	9
4	9	8	1	0	3	10
5	8	13	14	17	0	15
6	∞	∞	∞	∞	∞	0

$$A_5 = \{0, 1, 2, 3, 4, 5\}$$

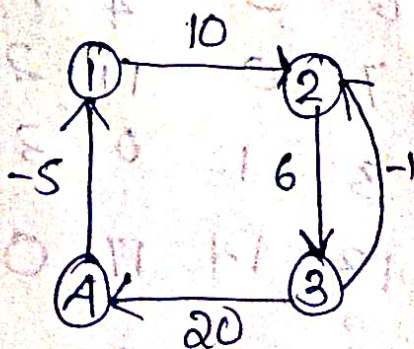
	1	2	3	4	5	6
1	0	5	6	9	8	7
2	1	0	5	4	7	2
3	8	7	0	11	2	9
4	9	8	1	0	3	10
5	8	13	14	17	0	15
6	∞	∞	∞	∞	∞	0

$$A_6 = \{0, 1, 2, 3, 4, 5, 6\}$$

	1	2	3	4	5	6
1	0	5	6	9	8	7
2	1	0	5	4	7	2
3	8	7	0	11	2	9
4	9	8	1	0	3	10
5	8	13	14	17	0	15
6	∞	∞	∞	∞	∞	0

- i) $1-6 : 7$
 ii) $4-1 : 9$

Q. Find the cost of 1-4 using all pair shortest path problem in dynamic programming



$$A_0 = \{0\}$$

	1	2	3	4
1	0	10	∞	∞
2	∞	0	6	∞
3	∞	-1	0	20
4	-5	∞	∞	0

$$A_1 = \{0, 1\}$$

	1	2	3	4
1	0	10	∞	∞
2	∞	0	6	∞
3	∞	-1	0	20
4	-5	5	∞	0

$$A_2 = \{0, 1, 2\}$$

	1	2	3	4
1	0	10	16	∞
2	∞	0	6	∞
3	∞	-1	0	20
4	-5	5	11	0

$$A_3 = \{0, 1, 2, 3\}$$

	1	2	3	4
1	0	10	16	36
2	21	0	6	26
3	∞	-1	0	20
4	-5	5	11	0

$$A_4 = \{0, 1, 2, 3, 4\}$$

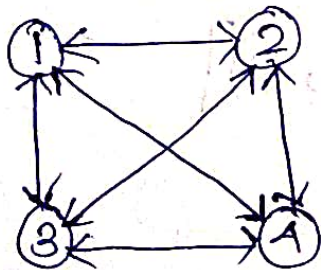
	1	2	3	4
1	0	10	16	36
2	21	0	6	26
3	15	-1	0	20
4	-5	5	11	0

path: $\{1, 2, 3, 4\}$

$$\text{Cost} = 10 + 6 + 20 = 36$$

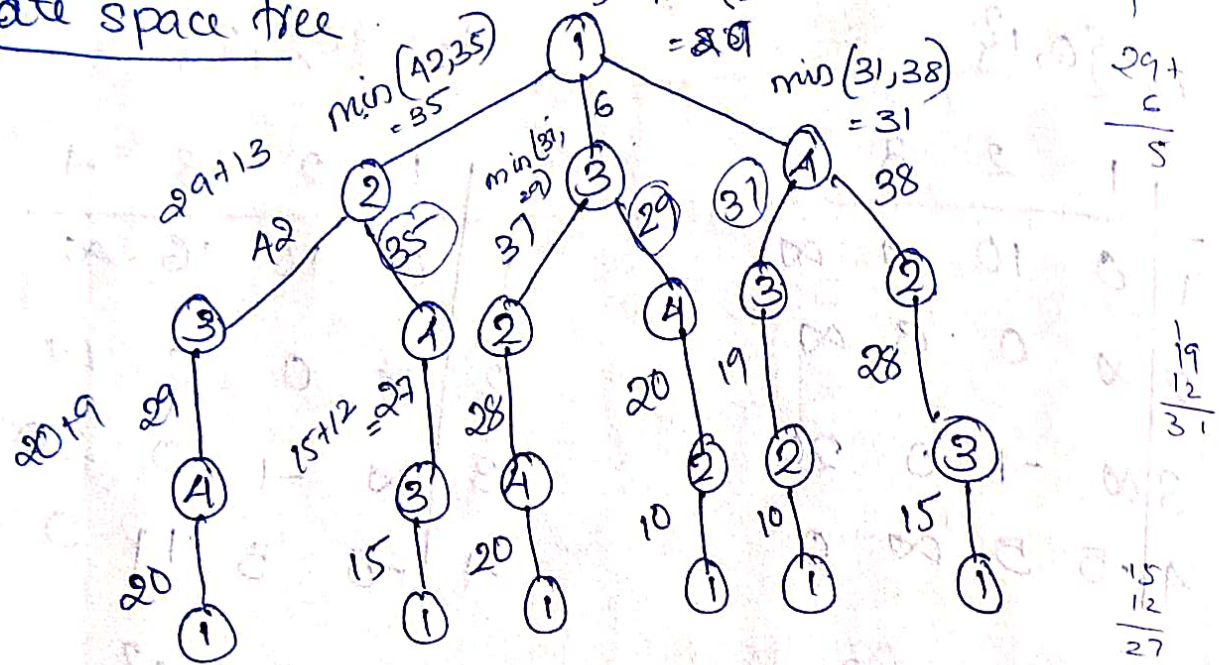
22/3/24
Friday

Travelling Salesman Problem is dynamic programming



	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

State space tree



Path: 1-3-4-2-1

Minimum cost = 35

Formula

$$g(i, s) = \min_{j \in s} [w(i, j) + g(j, s - j)]$$

$$g(1, \{2, 3, 4\}) = \min \begin{bmatrix} w[1, 2] + g(2, \{3, 4\}) \\ w[1, 3] + g(3, \{2, 4\}) \\ w[1, 4] + g(4, \{2, 3\}) \end{bmatrix}$$

i - starting node
s - intermediate
out of
j - element of s

$$a) \quad g(2, \{3, 4\}) = \min \begin{bmatrix} w[2, 3] + g(3, \{4\}) \\ w[2, 4] + g(4, \{3\}) \end{bmatrix}$$

$$\begin{aligned} a.1) \quad g(3, \{4\}) &= \min [w[3, 4] + g(4, \emptyset)] \\ &= w(3, 4) + w(4, 1) \\ &= 12 + 8 = \underline{\underline{20}} \end{aligned}$$

$$\begin{aligned}
 a.2) \quad g(4, \{3\}) &= \min [w[4,3] + g(3, \emptyset)] \\
 &= w(4,3) + w(3,1) \\
 &= 9 + 6 = \underline{\underline{15}}
 \end{aligned}$$

$$g(2, \{3,4\}) = \min \left[\begin{array}{c} 9 + 20 \\ 10 + 15 \end{array} \right] = \min(29, 25) = \underline{\underline{25}}$$

$$b) \quad g(3, \{2,4\}) = \min \left[\begin{array}{c} w[3,2] + g(2, \{4\}) \\ w[3,4] + g(4, \{2\}) \end{array} \right]$$

$$\begin{aligned}
 b.1) \quad g(2, \{4\}) &= \min [w[2,4] + g(4, \emptyset)] \\
 &= w(2,4) + w(4,1) \\
 &= 10 + 8 = \underline{\underline{18}}
 \end{aligned}$$

$$\begin{aligned}
 b.2) \quad g(4, \{2\}) &= \min [w[4,2] + g(2, \emptyset)] \\
 &= w(4,2) + w(2,1) \\
 &= 8 + 5 = \underline{\underline{13}}
 \end{aligned}$$

$$g(3, \{2,4\}) = \min \left[\begin{array}{c} 13 + 18 \\ 12 + 13 \end{array} \right] = \min(31, 25) = \underline{\underline{25}}$$

$$c) \quad g(4, \{2,3\}) = \min \left[\begin{array}{c} w[4,2], g(2, \{3\}) \\ w[4,3], g(3, \{2\}) \end{array} \right]$$

$$\begin{aligned}
 c.1) \quad g(2, \{3\}) &= \min [w[2,3], g(3, \emptyset)] \\
 &= w[2,3] + w[3,1] \\
 &= 9 + 6 = \underline{\underline{15}}
 \end{aligned}$$

$$\begin{aligned}
 c.2) \quad g(3, \{2\}) &= \min [w[3,2], g(2, \emptyset)] \\
 &= w[3,2] + w[2,1] \\
 &= 13 + 5 = \underline{\underline{18}}
 \end{aligned}$$

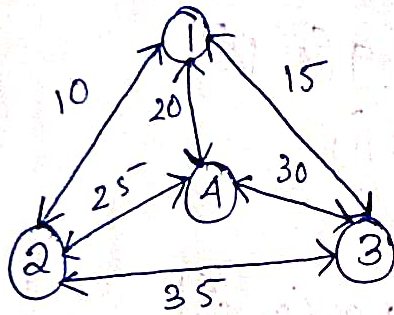
$$g(4, \{2, 3\}) = \min [9+18]$$

$$= \min (23, 27) = \underline{23}$$

$$g(1, \{2, 3, 4\}) = \min \begin{bmatrix} 10+25 \\ 15+25 \\ 20+23 \end{bmatrix}$$

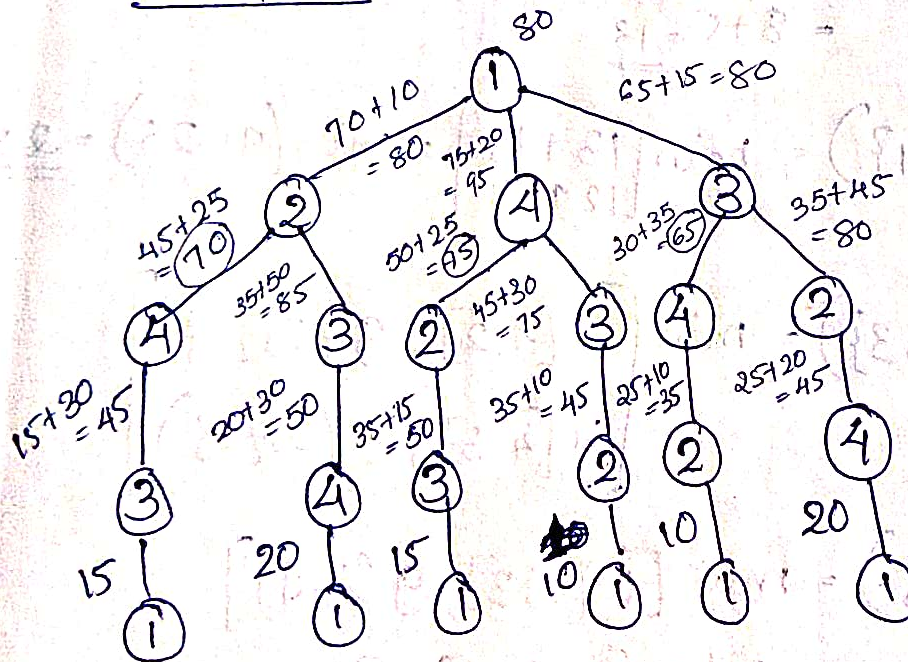
$$= \min (35, 40, 43) = \underline{35}$$

Q.



	1	2	3	4
1	0	10	15	20
2	10	0	35	25
3	15	35	0	30
4	20	25	30	0

State space tree



$$\frac{35}{15} = 0$$

$$\frac{45+25}{70} = 1$$

$$\frac{35}{45} = 0$$

$$\frac{75+20}{95} = 1$$

$$\frac{65+15}{80} = 1$$

Minimum cost : 80

Path : 1-2-4-3-1 or 1-3-4-2-1

Formula

$$g(i, s) = \min_{j \in s} [w(i, j) + g(j, \{s - j\})]$$

$$g(1, \{2, 3, 4\}) = \min \begin{bmatrix} w[1, 2] + g(2, \{3, 4\}) \\ w[1, 3] + g(3, \{2, 4\}) \\ w[1, 4] + g(4, \{2, 3\}) \end{bmatrix}$$

$$a). g(2, \{3, 4\}) = \min \begin{bmatrix} w[2, 3] + g(3, \{4\}) \\ w[2, 4] + g(4, \{3\}) \end{bmatrix}$$

$$\begin{aligned} a.1) g(3, \{4\}) &= \min [w[3, 4] + g(4, \emptyset)] \\ &= w(3, 4) + w(4, 1) \\ &= 30 + 20 = \underline{50} \end{aligned}$$

$$\begin{aligned} a.2) g(4, \{3\}) &= \min [w[4, 3] + g(3, \emptyset)] \\ &= w(4, 3) + w(3, 1) \\ &= 30 + 15 = \underline{45} \end{aligned}$$

$$g(2, \{3, 4\}) = \min \begin{bmatrix} 35 + 50 \\ 25 + 45 \end{bmatrix} = \min(85, 70) = \underline{70}$$

$$b) g(3, \{2, 4\}) = \min \begin{bmatrix} w[3, 2] + g(2, \{4\}) \\ w[3, 4] + g(4, \{2\}) \end{bmatrix}$$

$$\begin{aligned} b.1) g(2, \{4\}) &= \min [w[2, 4] + g(4, \emptyset)] \\ &= w(2, 4) + w(4, 1) \\ &= 25 + 20 = \underline{45} \end{aligned}$$

$$\begin{aligned} b.2) g(4, \{2\}) &= \min [w[4, 2] + g(2, \emptyset)] \\ &= w(4, 2) + w(2, 1) \\ &= 25 + 10 = \underline{35} \end{aligned}$$

$$g(3, \{2, 4\}) = \min \begin{bmatrix} 35 + 45 \\ 30 + 35 \end{bmatrix} \\ = \min(80, 65) = \underline{65}$$

$$c) g(4, \{2, 3\}) = \min \begin{bmatrix} w(4, 2) + g(2, \{3\}) \\ w(4, 3) + g(3, \{2\}) \end{bmatrix}$$

$$c1) g(2, \{3\}) = \min [w[2, 3] + g(3, \emptyset)] \\ = w(2, 3) + w(3, 1) \\ = 35 + 15 = \underline{50}$$

$$c2) g(3, \{2\}) = \min [w[3, 2] + g(2, \emptyset)] \\ = w[3, 2] + w[2, 1] \\ = 35 + 10 = \underline{45}$$

$$g(4, \{2, 3\}) = \min \begin{bmatrix} 25 + 50 \\ 30 + 45 \end{bmatrix} = \min(75, 75) = \underline{75}$$

$$g(1, \{2, 3, 4\}) = \min \begin{bmatrix} 10 + 70 \\ 15 + 65 \\ 20 + 45 \end{bmatrix} = \min(80, 80, 65) = \underline{65}$$

$$g(1, \{2, 3, 4\}) = \min \begin{bmatrix} 10 + 70 \\ 15 + 65 \\ 20 + 45 \end{bmatrix} \\ = \min(80, 80, 65) \\ = \underline{65}$$

$$\underline{65}$$

26/3/24
Tuesday

Backtracking

It is one of the problem solving algorithm design approach. Backtracking build incremental solution until it finds dead end. If a deadend found, it backtracks to the initial position to get another option for solution.

Application

- N-queens problem
- graph coloring
- sum of subsets
- 0/1 knapsack problem

Algorithm Backtrack(k)

{ for each $x[k] \in T(x[1], \dots, x[k-1])$

{ if $B_k(x[1], x[2], \dots, x[k]) \neq 0$
then

{ if $x[1], x[2], \dots, x[k]$ is a path to an
answer node

then write $(x[1:k])$

if $(k < n)$ then Backtrack($k+1$)

{ } { }

$(x[1:n])$ = solution vector

$T()$ - generalizing function

$B()$ - Backtrack

N-Queens problem

Given a problem to place n -queens on a $n \times n$ chessboard, the condition is no two queens are in same row, same column or diagonal.

eg: 4×4

	1	2	3	4
1	Q_1			
2			Q_2	
3				
4				

To place Q_1, Q_2, Q_3, Q_4 in 4×4

Q_3 can't be placed. it violates the condition. So backtrack. place Q_1 in c_2 and Q_2 in c_4 .

	1	2	3	4
1		Q_1		
2				Q_2
3	Q_3			
4			Q_4	

(2 4 1 3) column no.

1	2	3	4
Q_3	Q_1	Q_4	Q_2

mirror: 3 1 4 2 is also a sol.

To check diagonal

upper left

upper right

	1	2	3	4
1	X			
2	X	X		X
3		X	X	
4				X

(1,1) (2,2) (3,3) (4,4)

(2,1) (3,2) (4,3)

Condition to check the formation of diagonal from upper left to right.

∴ every element on the same diagonal that goes from the upper left to the right has row - column = same value.

Every element on the same diagonal that goes from the upper right to lower left has the same row + column value.

$$\begin{array}{l} \text{row} - \text{col} \\ i - j = k - l \\ (3 - 2) = (4 - 3) \\ 1 = 1 \end{array}$$

$$\Downarrow \\ |j - l| = |i - k|$$

$$\begin{array}{l} \text{row} + \text{col} \\ i + j = k + l \\ (2 + 4) = (3 + 3) \\ 6 = 6 \end{array}$$

Solution for 8-queens problem

	1	2	3	4	5	6	7	8
1	Q ₁							
2			Q ₂					
3					Q ₃			
4		Q ₄						
5				Q ₅				
6						Q ₆		
7							Q ₇	
8								

Can't place Q₈.
Backtrack.

	1	2	3	4	5	6	7	8
1		Q ₁						
2				Q ₂				
3	Q ₃							
4			Q ₄					
5					Q ₅			
6							Q ₆	
7								
8						Q ₇		

Can't place Q₈:
Back track.

	1	2	3	4	5	6	7	8
1			Q ₁					
2	Q ₂							
3				Q ₃				
4		Q ₄						
5					Q ₅			
6							Q ₆	
7								
8						Q ₇		

$$(2-1) = (1-0)$$

$$(3-2) = (2-1)$$

$$(4-3) = (3-2)$$

	1	2	3	4	5	6	7	8
1				Q ₁				
2	Q ₂							
3			Q ₃					
4					Q ₄			
5		Q ₅						
6						Q ₆		
7								Q ₇
8								

Group 8 not suitable

	1	2	3	4	5	6	7	8
1								
2								
3								
4								
5								
6								
7								
8								

Algorithm nqueen(k, n)

{
 for i = 1 to n do

 {
 place(k, i) then

 {
 $x[k] = i$

 if (k = n) then write $x[1 \dots n]$

 else

 nqueen(k+1, n)

 place(k, i)

return true value if a queen can be placed
in kth row ith column. else return false. $x[]$
is an array. $\text{abs}(x)$ returns absolute value of x .

{
 for j = 1 to k-1

 do

 {

 if $x[j] = i$ // same column

 or

$\text{abs}[x[j] - i] = \text{abs}(j - k)$

 // same diagonal

 then return false;

 return true;

}

	1	2	3	4	5	6	7	8
1					Q ₁			
2	Q ₂							
3			Q ₃					
4						Q ₄		
5		Q ₅						
6				Q ₆				
7							Q ₇	
8								

	1	2	3	4	5	6	7	8
1						Q ₁		
2	Q ₂							
3			Q ₃					
4					Q ₄			
5		Q ₅						
6				Q ₆				
7						Q ₇		
8								

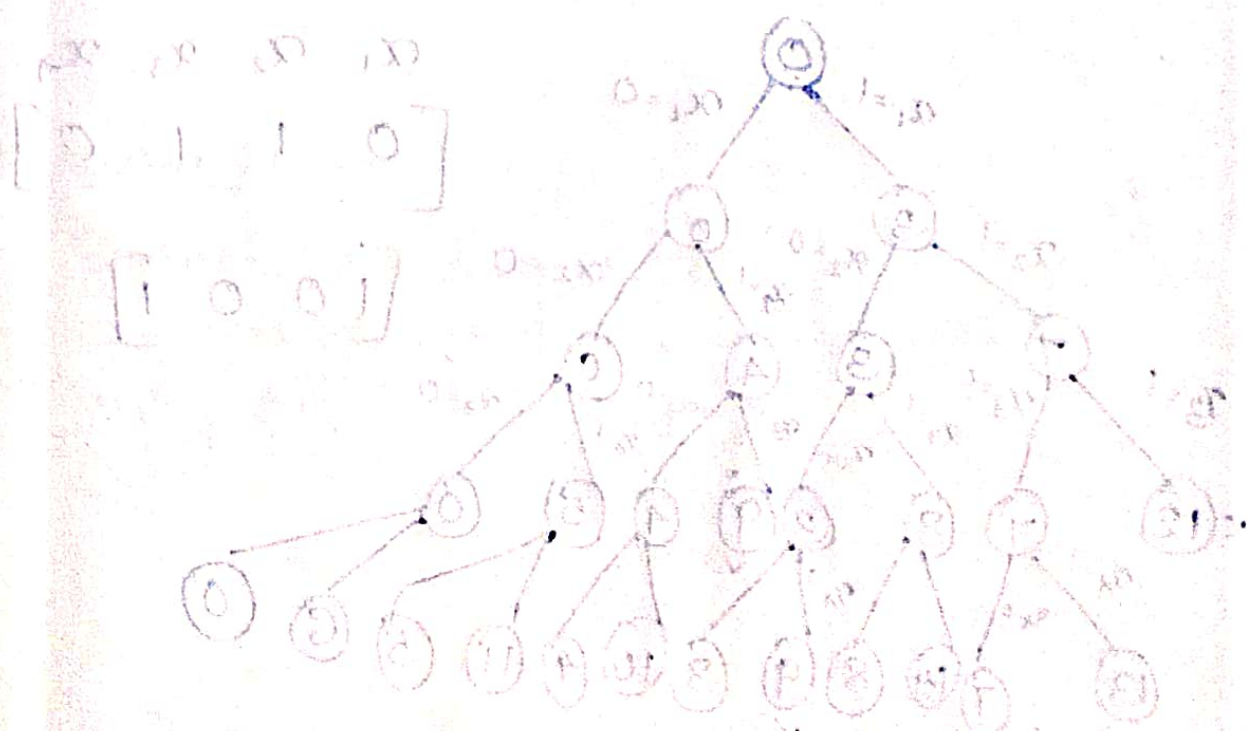
	1	2	3	4	5	6	7	8
1							Q ₁	
2	Q ₂							
3			Q ₃					
4					Q ₄			
5		Q ₅						
6				Q ₆				
7						Q ₇		
8								Q ₈

	1	2	3	4	5	6	7	8
	Q ₂	Q ₅	Q ₃	Q ₆	Q ₄	Q ₇	Q ₁	Q ₈

placed all queens

A hand-drawn 8x8 grid with columns and rows labeled 1 to 8. Queens are placed at positions (1,8), (2,1), (3,3), (4,6), (5,5), (6,7), (7,2), and (8,4).

$\Delta P = 10$
 $P = 20$



27/3/21
Wednesday

sum of subset

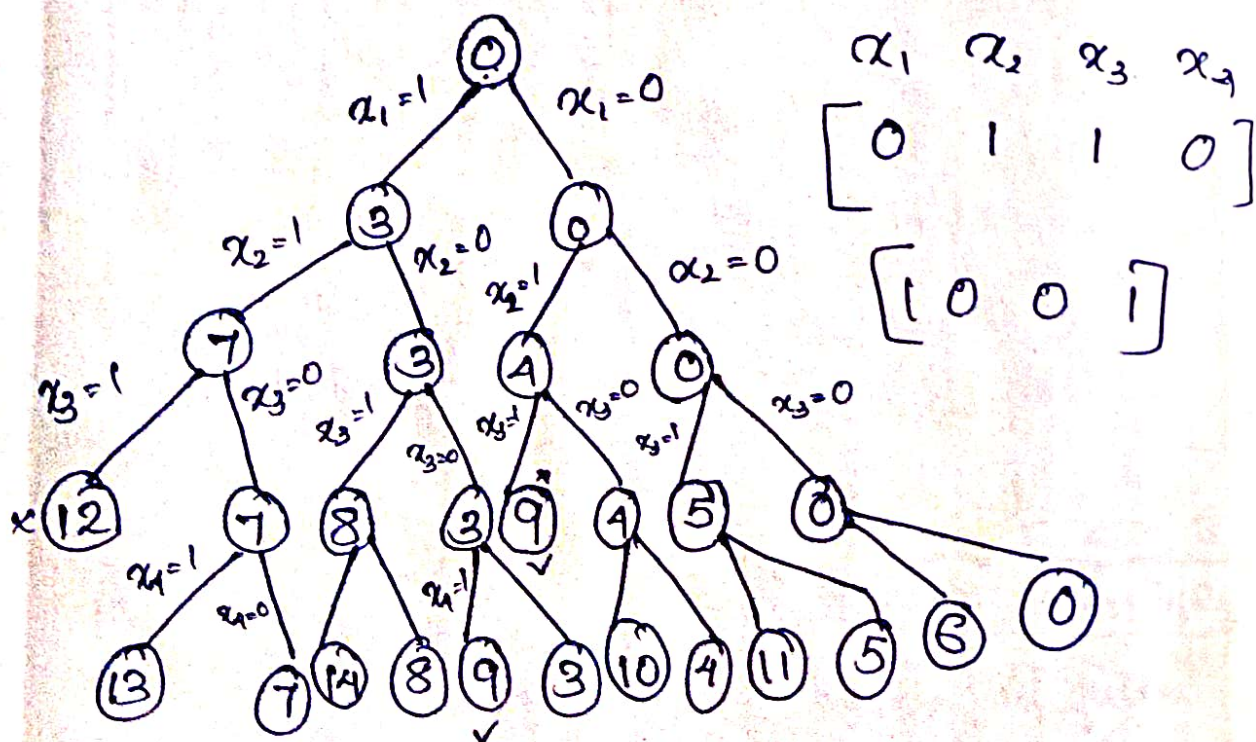
In the subset sum problem, we have to find a subset S' of the given set.

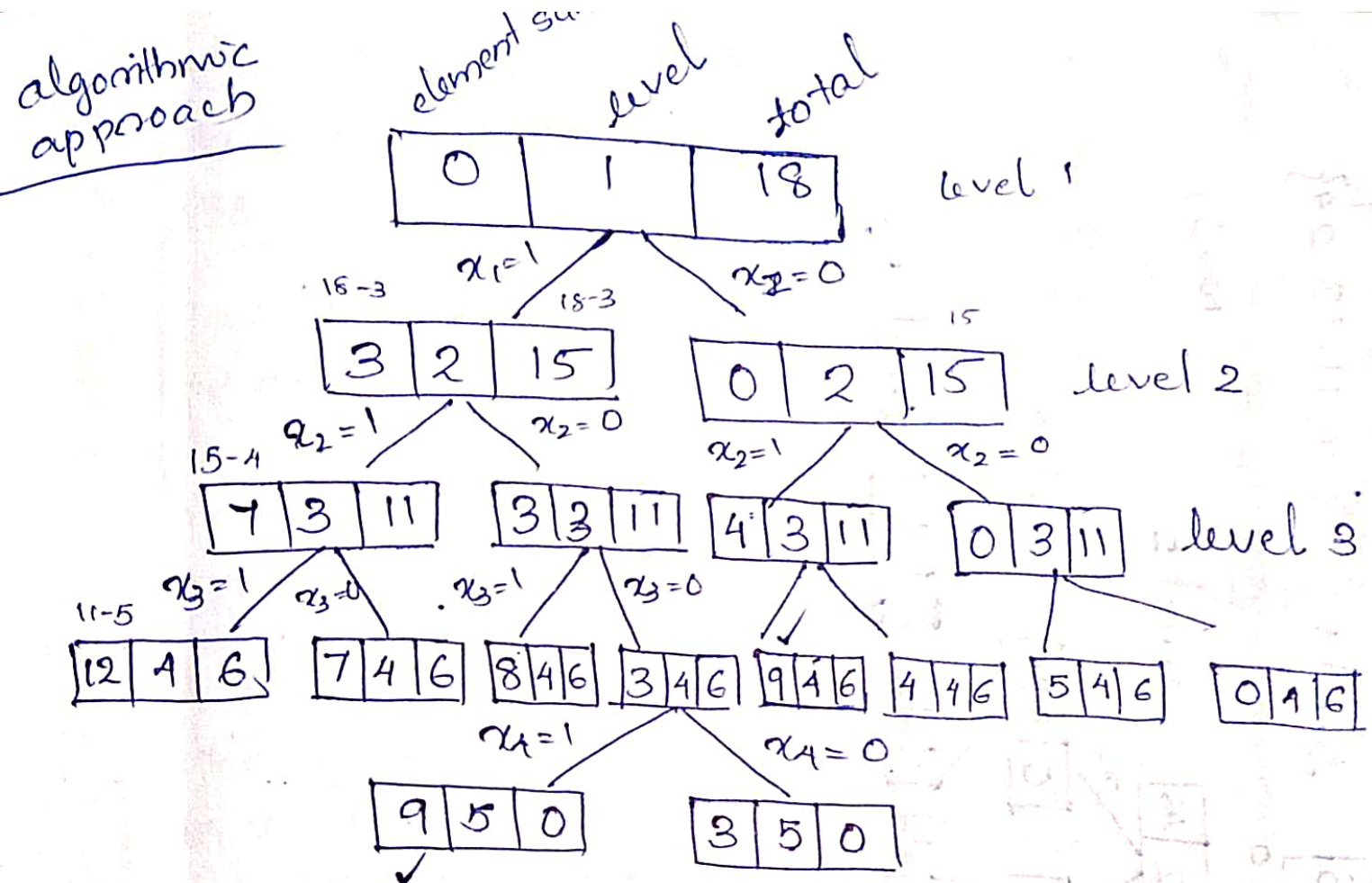
$S = \{x_1, x_2, x_3, \dots, x_n\}$ where the elements of the set S are n positive integers in such a manner that $x' \in S$ and sum of the elements $S = \text{same positive integer } S$.

eg: Given a set $S = \{3, 4, 5, 6\}$ and $x = 9$ obtain the subset sum using backtracking approach.

$$S = \{x_1, x_2, x_3, x_4\} = \{3, 4, 5, 6\}$$

$$M = 9 / x = 9$$





Q. Solve the following using subset sum problem where elements are $S = \{7, 11, 13, 24\}$ $M = 31$. Solve using algorithmic approach.

⊗ Seminar

Branch and bound

depth first

using knapsack problem

$$\frac{18}{3} = 6$$

$$\frac{45}{37} = 1.216$$

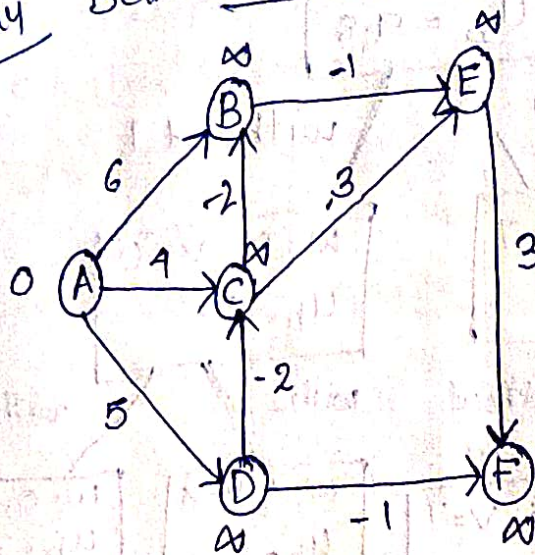
$$\frac{55}{48} = 1.145$$

$$\frac{37}{24} = 1.541$$

3/4/24
Wednesday

Bellman Ford Algorithm

6 vertices
So $6-1=5$
iterations.



Relaxing,

$$\text{if } (d[u] + c(u, v) < d[v])$$

$$d[v] = d[u] + c(u, v)$$

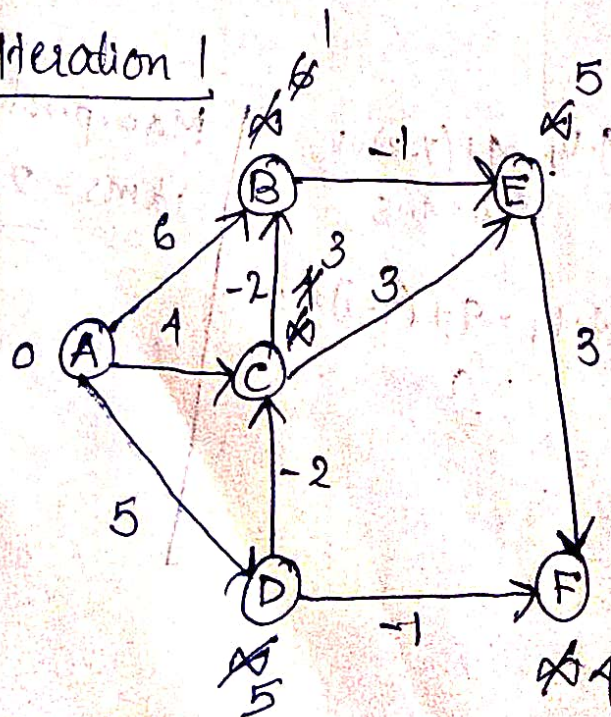
Edge list

(A, B), (A, C), (A, D)

(B, E), (C, E), (D, C)

(D, F), (E, F), (C, B)

Iteration 1



$$(A, B) - 0 + 6 < \infty$$

$$\text{So, } B = 6$$

$$(B, E) - 6 + (-1) < \infty$$

$$5 < \infty$$

$$\text{So } E = 5$$

$$(A, C) - 0 + 4 < \infty$$

$$\text{So, } C = 4$$

$$(A, D) - 0 + 5 < \infty$$

$$\text{So } D = 5$$

$$(A, E) - 4 + 3 \neq 5$$

$$\text{So, } E = 5$$

$$(D, C) = 5 + (-2) < 4$$

$$3 < 4$$

$$\text{So, } C = 3$$

$$(D, F) = 5 + (-1) < \infty$$

$$4 < \infty$$

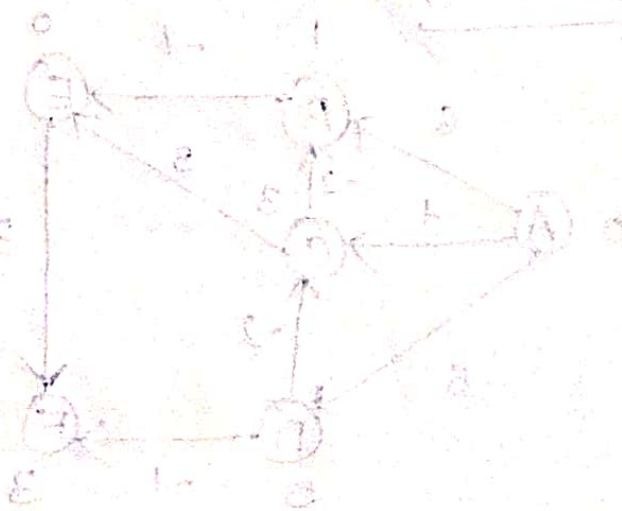
$$\text{So, } F = 4$$

$$(E, F) = 5 + 3 \neq 4$$

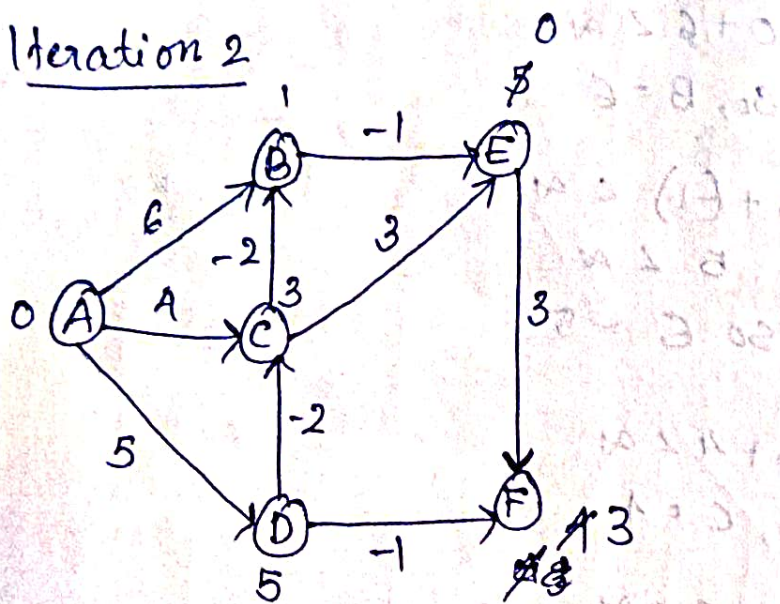
$$(C, B) = 3 + (-2) < 6$$

$$= 1 < 6$$

$$\text{So, } B = 1$$



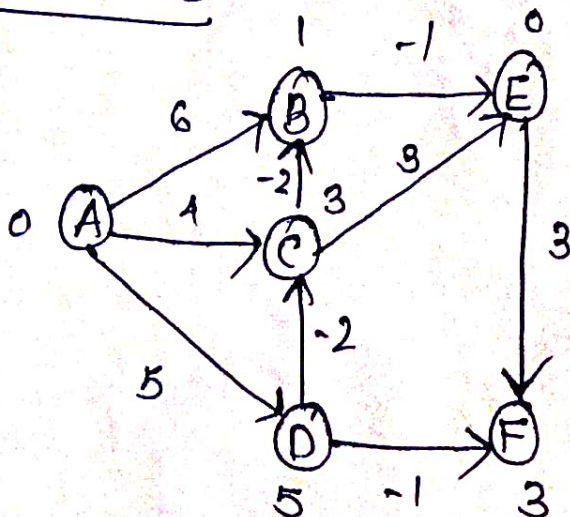
Iteration 2



Edge list

(A, B) 1	(A, C) 3	(A, D) 5
(B, E) 0	(C, E) 0	(D, E) 3
(D, F) 4	(E, F) 3	(C, B) 1

Iteration 3



Edge list

(A, B) 1	(A, C) 3	(A, D) 5	(B, E) 0
(C, E) 0	(D, C) 3	(D, F) 3	(E, F) 3
			(C, B) 1

No change in cost.

So o/p is

A - 0

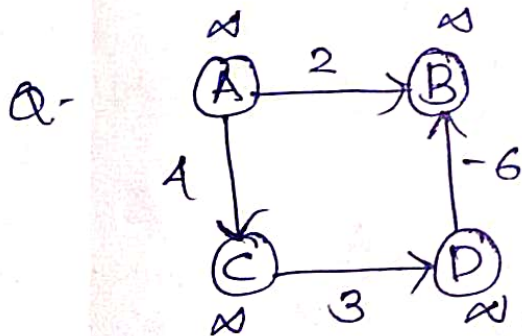
F - 3

B - 1

C - 3

D - 5

E - 0

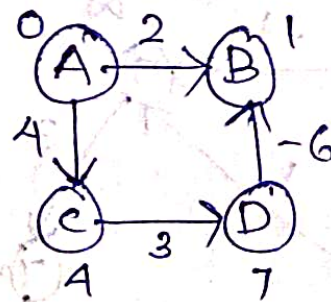
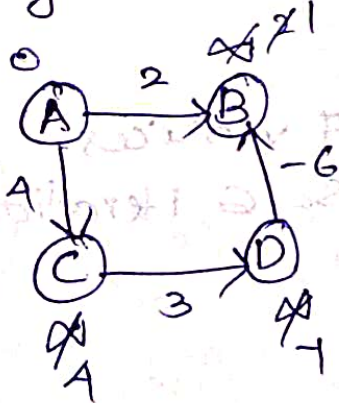


4 vertices.

So, 3 iteration.

Iteration 1

Edge list: (A, B) (A, C) (C, D) (D, B)



Iteration 2

(A, B) (A, C) (C, D) (D, B)

1

A

7

1

No change in cost.

So o/p is: (F, C) (G, D) (H, E) (I, B)

A: 0, B: 1, C: 4, D: 7

Time complexity

$$O(E |V|-1)$$

$$O(EV)$$

$$O(n^2)$$

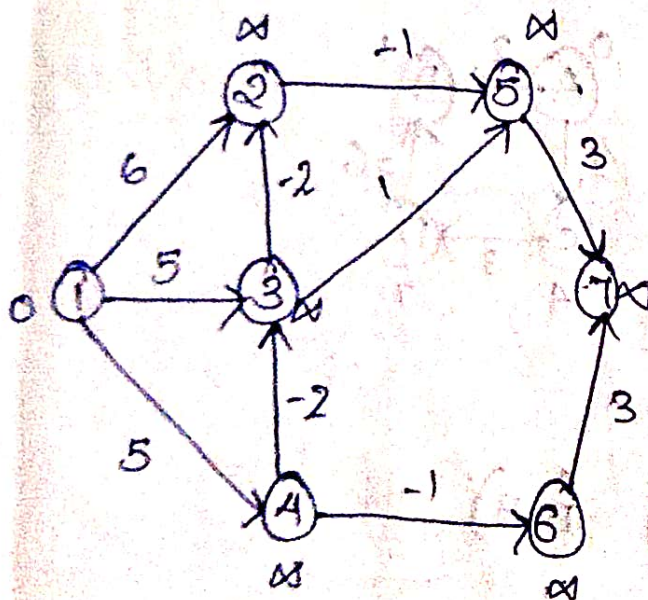
TC of complete graph

$$E = \frac{V(V-1)}{2}$$

$$\left[\frac{n(n-1)}{2} (n-1) \right]$$

$$= \underline{\underline{O(n^3)}}$$

Q.

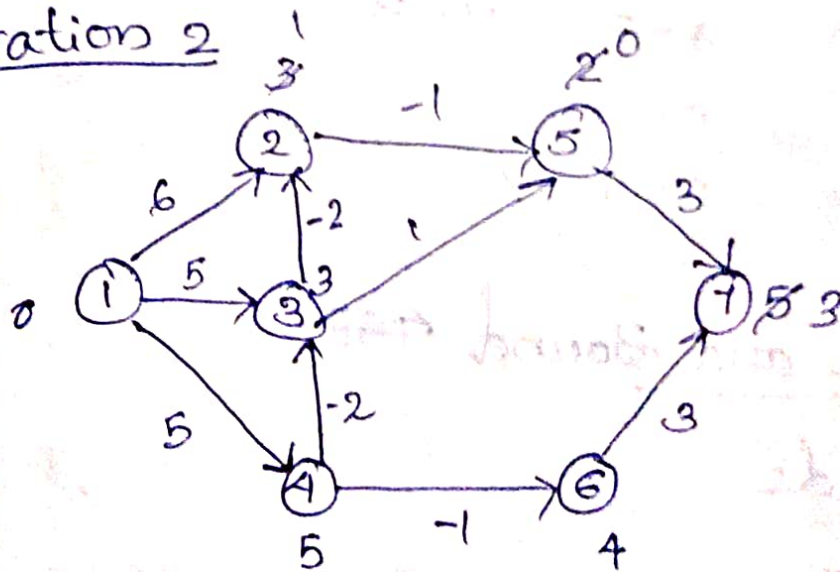
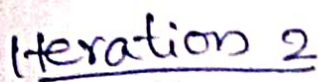


7 vertices.
So 6 iteration

Iteration 1

Edges: (1,2) (1,3) (1,4) (3,2) (4,3)

(2,5) (3,5) (4,6) (5,7) (6,7)

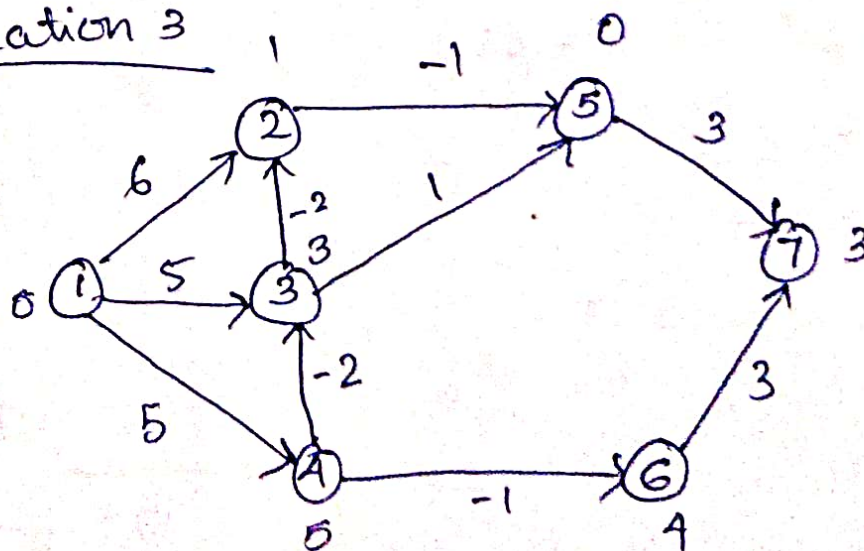


edges :

edges :

$\begin{pmatrix} 1,2 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 1,3 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 1,4 \\ 5 \end{pmatrix}$	$\begin{pmatrix} 3,2 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 4,3 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 2,5 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3,5 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 4,6 \\ 1 \end{pmatrix}$
$\begin{pmatrix} 5,7 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 6,7 \\ 3 \end{pmatrix}$						

Iteration 3



$\binom{1,2}{1}$ $\binom{1,3}{3}$ $\binom{1,4}{5}$ $\binom{2,2}{1}$ $\binom{1,3}{3}$ $\binom{2,5}{0}$
 $\binom{3,5}{0}$ $\binom{4,6}{1}$ $\binom{5,7}{3}$ $\binom{6,7}{3}$

cf :

1-0	4-5	7-3
2-1	5-0	
3-3	6-4	

7/1/24
Thursday

11/1/24
day, Branch and Bound ~~for~~

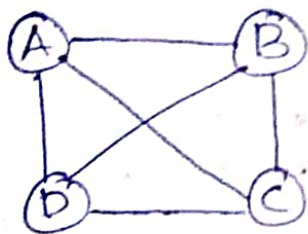
live node

e-node

pouring

dead node

Branch and bound TSP



	A	B	C	D	Min
A	∞	10	5	3	3
B	8	∞	9	7	7
C	1	6	∞	9	1
D	2	3	8	∞	2
					<u>13</u>

Reduced cost = 13

- subtract the min value from each row.

	A	B	C	D
A	∞	7	2	0
B	1	∞	2	0
C	0	5	∞	8
D	0	1	6	∞

To make zero in each row

Step 2: Column reduction

	A	B	C	D
A	∞	7	2	0
B	1	∞	2	0
C	0	5	∞	8
D	0	1	6	∞
	0	1	2	0

'upper' = A

Reduced cost

$$RC = 13 + 3 = \underline{\underline{16}}$$

Step 3: Reduce each value in the column with min to make zero

Matrix A

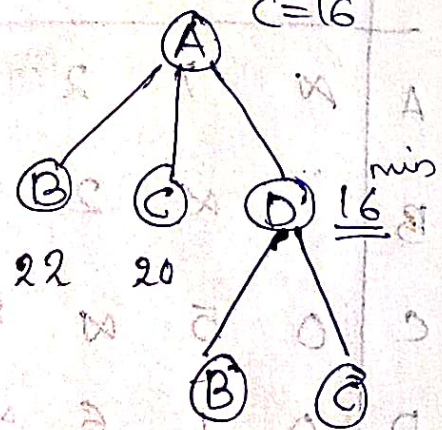
	A	B	C	D
A	∞	6	0	0
B	1	∞	0	0
C	0	4	∞	8
D	0	0	4	∞

$$\infty - \infty = \infty$$

$$4 - 1 = 3$$

Cost of A to B

	A	B	C	D
A	∞	∞	∞	∞
B	1	∞	0	0
C	0	∞	∞	8
D	0	∞	4	∞



$$TC = C(A, B) + RC + RC'$$

$$= 6 + 16 + 0$$

$$= \underline{\underline{22}}$$

A to C

Row reduction

	A	B	C	D	
A	∞	∞	∞	∞	0 +
B	1	∞	∞	0	0
C	∞	4	∞	8	4
D	0	0	∞	∞	0
					<u>4</u>

subtract only here.

column reduction

	A	B	C	D
A	∞	∞	∞	∞
B	1	∞	∞	0
C	∞	0	∞	4
D	0	0	∞	∞
	0	0	0	0

$$RC' = 4 + 0 = 4$$

$$\begin{aligned}
 TC &= C(A, C) + RC + RC' \\
 &= 0 + 16 + 4 \\
 &= \underline{\underline{20}}
 \end{aligned}$$

reduction was as

	A	B	C	D
A	∞	∞	∞	∞
B	1	∞	∞	0
C	∞	0	∞	4
D	0	0	∞	∞
	0	0	0	0

A to D

	A	B	C	D	ms
A	1	1	1	1	0
B	1	1	0	1	0
C	0	1	1	1	0
D	1	0	1	1	0
	0	0	0	0	0

$$\begin{aligned}
 TC &= C(A, D) + RC + RC' \\
 &= 0 + 16 + 0 \\
 &= \underline{\underline{16}}
 \end{aligned}$$

Use A to last A-D matrix to compute

D to B

	A	B	C	D	ms
A	1	1	1	1	0
B	1	1	0	1	0
C	0	1	1	1	0
D	1	1	1	1	0
	0	0	0	0	0

no row reduction

$$TC = C(D, B) + RC + RC'$$

$$= 0 + 16 + 0$$

D to C

	A	B	C	D	
A	∞	∞	∞	∞	0
B	1	∞	∞	∞	1
C	0	4	∞	∞	0
D	∞	∞	∞	∞	0
					<u>1</u>

row reduction

	A	B	C	D
A	∞	∞	∞	∞
B	0	∞	∞	∞
C	0	4	∞	∞
D	∞	∞	∞	∞
	0	4	0	0

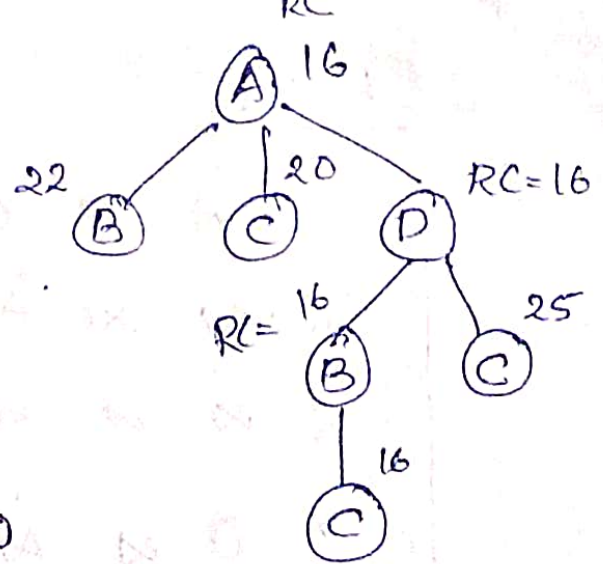
$$RC' = 1 + 4 = 5$$

column reduction.

$$TC = C(D, C) + RC + RC'$$

$$= 0 + 16 + 5$$

$$= \underline{\underline{25}}$$



B to C

	A	B	C	D	Min
A	∞	∞	∞	∞	0
B	∞	∞	∞	∞	0
C	0	∞	∞	∞	0
D	∞	∞	∞	∞	0
	0	0	0	0	0

$$RC' = 0$$

$$T(B, C) + RC + RC'$$

$$= 0 + 16 + 0 = \underline{\underline{16}}$$

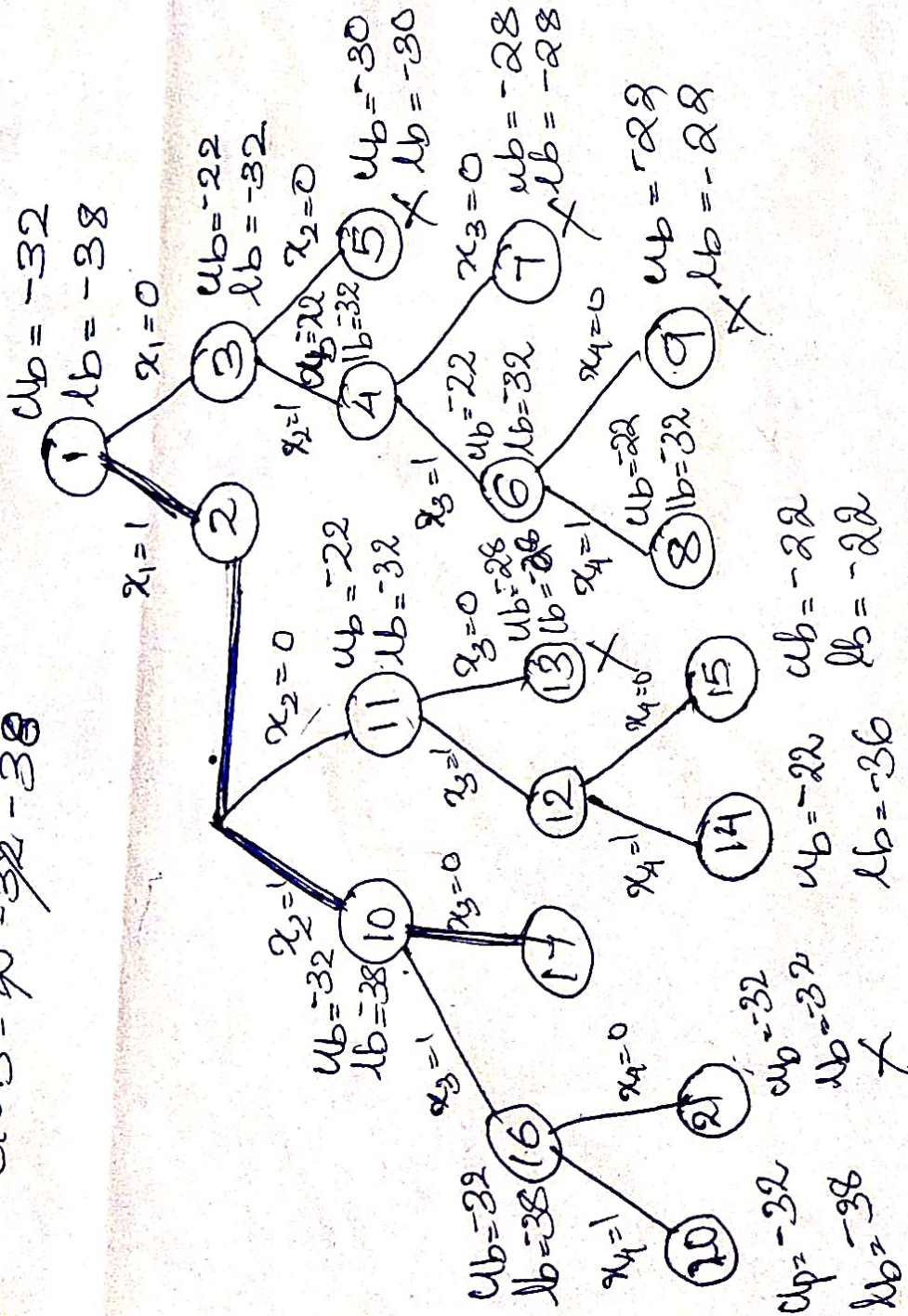
Upper = ∞

$\infty > 16$, so, upper = 16

Path: ADBC

0/1 knapsack problem using LIFO

$$GUB = -32 - 38$$



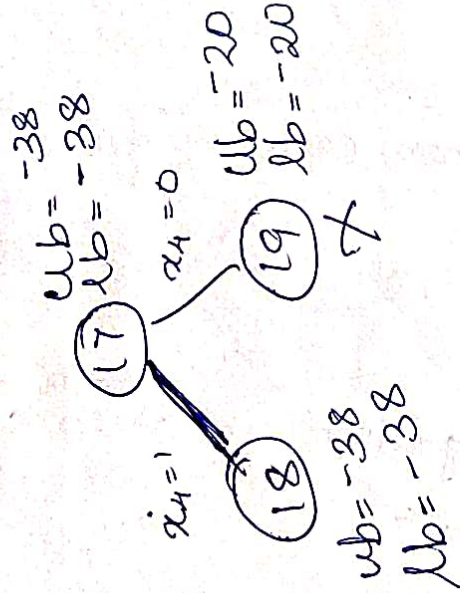
$$ub = P_i x_i$$

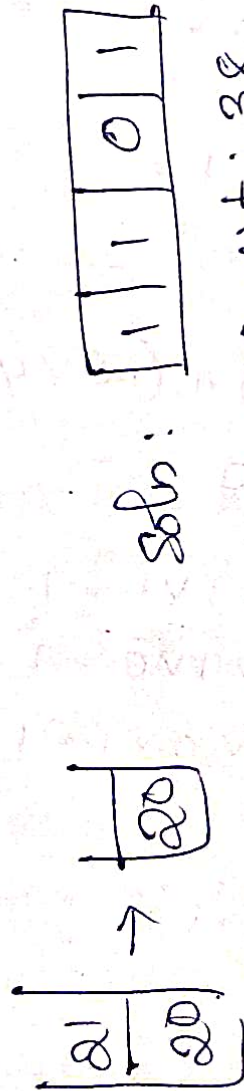
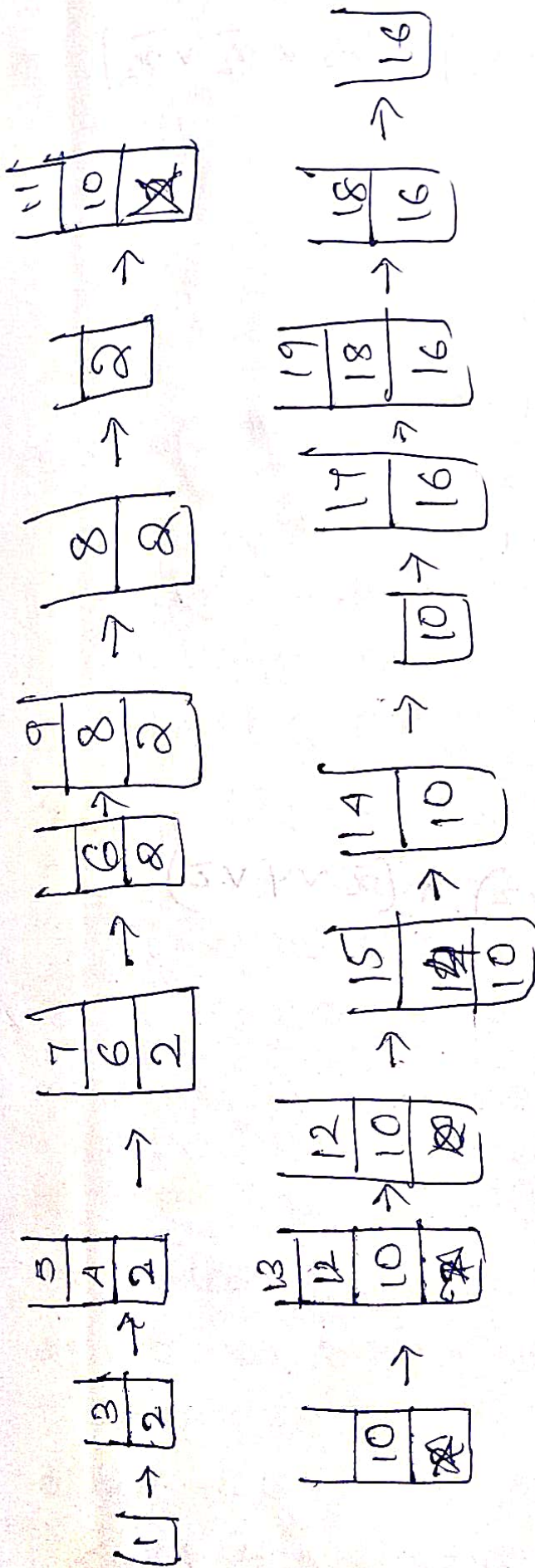
$$lb = P_i f_i$$

$$M=15$$

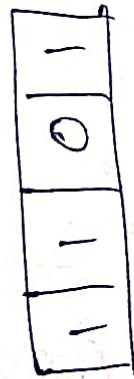
P	-10	-10	-12	-18
W	2	4	6	9

1. $ub < GUB \Rightarrow$ update GUB
 2. $lb > GUB \Rightarrow$ kill node





soln:



profit = 38
wt = 15

11/4/24
Thursday

Prove that the following problem is satisfiable or not.

$$(\alpha_1 \vee \alpha_2 \vee \bar{\alpha}_3) \wedge (\bar{\alpha}_1 \vee \alpha_3 \vee \alpha_4) \wedge (\bar{\alpha}_0 \vee \bar{\alpha}_2 \vee \bar{\alpha}_3) \\ \wedge (\alpha_1 \vee \alpha_2 \vee \bar{\alpha}_3)$$

Proof

Assume $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be 0

$$\alpha_1 \vee \alpha_2 \vee \bar{\alpha}_3 \Rightarrow 0 \vee 0 \vee 1 = 1$$

$$\bar{\alpha}_1 \vee \alpha_3 \vee \alpha_2 \Rightarrow 1 \vee 0 \vee 0 = 1$$

$$\bar{\alpha}_1 \vee \bar{\alpha}_2 \vee \bar{\alpha}_3 \Rightarrow 1 \vee 1 \vee 1 \Rightarrow 1$$

$$\alpha_1 \vee \alpha_2 \vee \bar{\alpha}_3 \Rightarrow 0 \vee 0 \vee 1 = 1$$

$$1 \wedge 1 \wedge 1 \wedge 1 = 1$$

Satisfiable.

$$Q. S = (\alpha \vee \gamma \vee \bar{z}) \wedge (\bar{\alpha} \vee \gamma \vee z) \wedge (\bar{\alpha} \vee \bar{\gamma} \vee \bar{z})$$

$$\alpha=1, \gamma=1, z=0$$

$$\alpha \vee \gamma \vee \bar{z} = 1 \vee 1 \vee 1 = 1$$

$$\bar{\alpha} \vee \gamma \vee z = 0 \vee 1 \vee 0 = 1$$

$$\bar{\alpha} \vee \bar{\gamma} \vee \bar{z} = 0 \vee 0 \vee 1 = 1$$

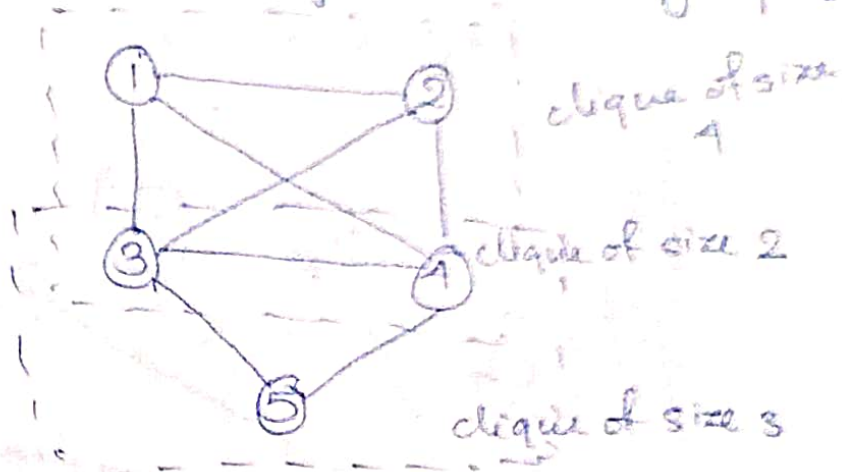
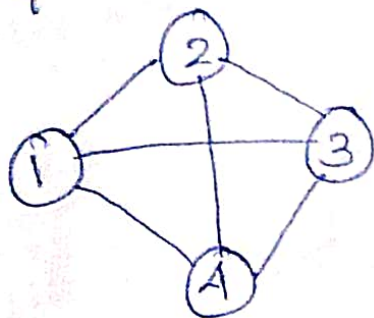
$$1 \wedge 1 \wedge 1 = 1$$

satisfiable.

Algo
wednesday

clique

clique is the complete subgraph of a graph.



a. show that clique is NP complete

If a problem is NP means which can't be solved in polynomial time. But it is verifiable in polynomial time is known as non-deterministic polynomial or NP class problem.

Step 1 : Clique is NP problem

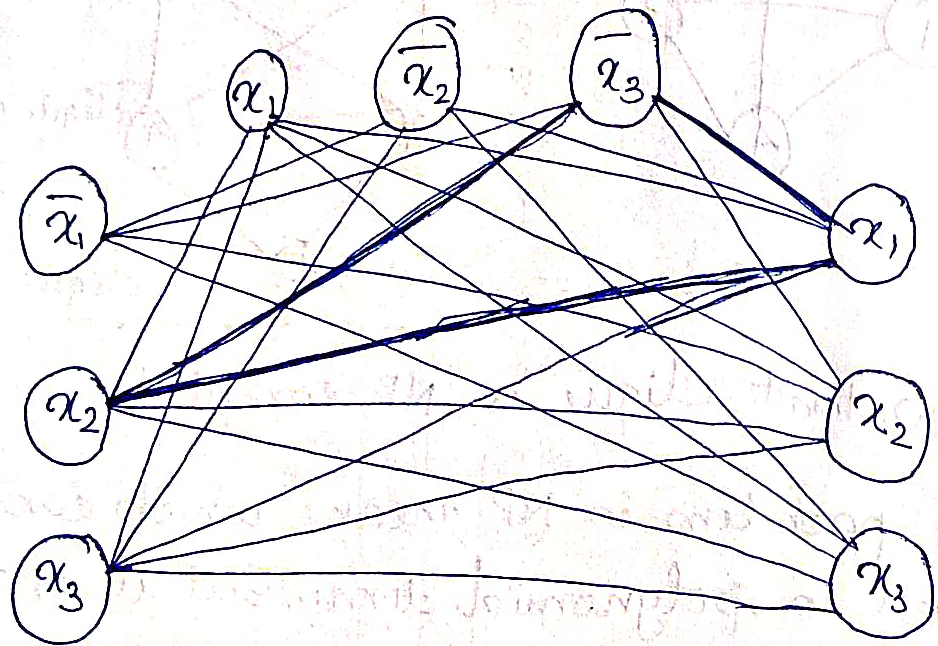
Step 2 : A non NPC (3CNF SAT) problem can be reduced to clique decision problem

COP. 3CNF SAT ~~is~~ as COP
reduced to

Rules for constructing clique from satisfiability problem.

1. Vertex of same clause should not be connected.
2. We can connect vertex of one clause with the vertex of another clause.
3. We should not connect a vertex in one clause with the negation vertex in another clause.

eg: $\phi = (\alpha_1 \vee \bar{\alpha}_2 \vee \bar{\alpha}_3) \wedge (\bar{\alpha}_1 \vee \alpha_2 \vee \alpha_3) \wedge (\alpha_1 \vee \alpha_2 \vee \alpha_3)$



clique:

Above graph contain clique of size 3 and 2,

$k=3$

for ex: $\alpha_1, \alpha_2, \alpha_3$ (take the max-size clique, here 3)

$$\phi = (\alpha_1 \vee \bar{\alpha}_2 \vee \bar{\alpha}_3) \wedge (\bar{\alpha}_1 \vee \alpha_2 \vee \alpha_3) \wedge (\alpha_1 \vee \alpha_2 \vee \alpha_3)$$

$$= (1 \vee 0 \vee 1) \wedge (0 \vee 1 \vee 1) \wedge (1 \vee 1 \vee 1)$$

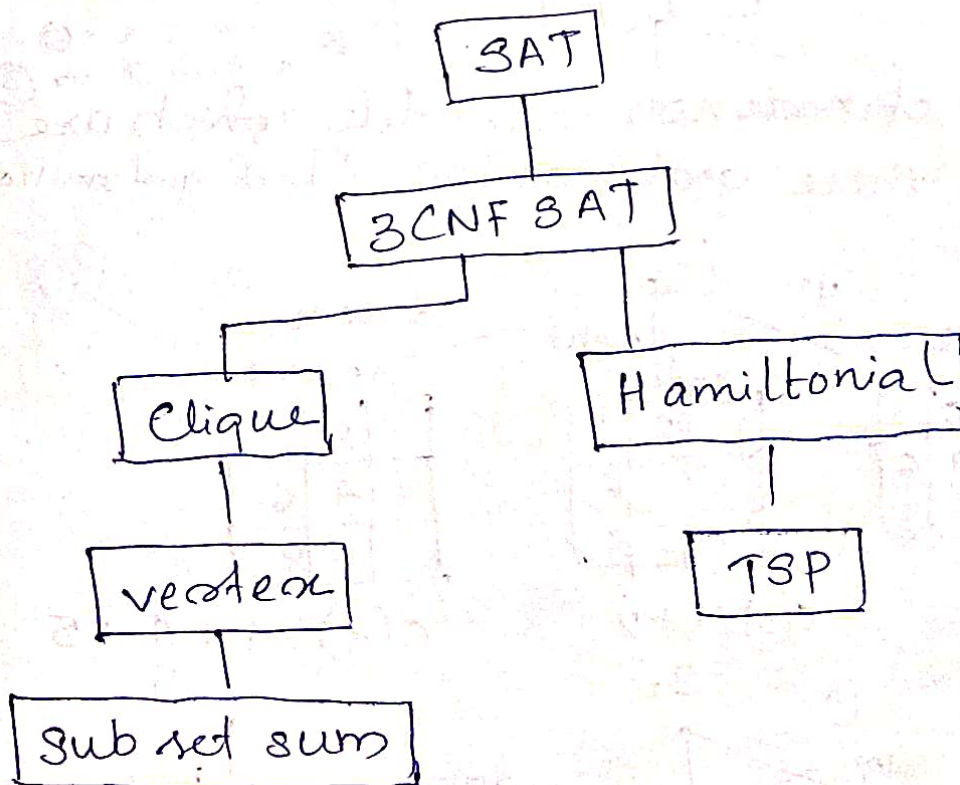
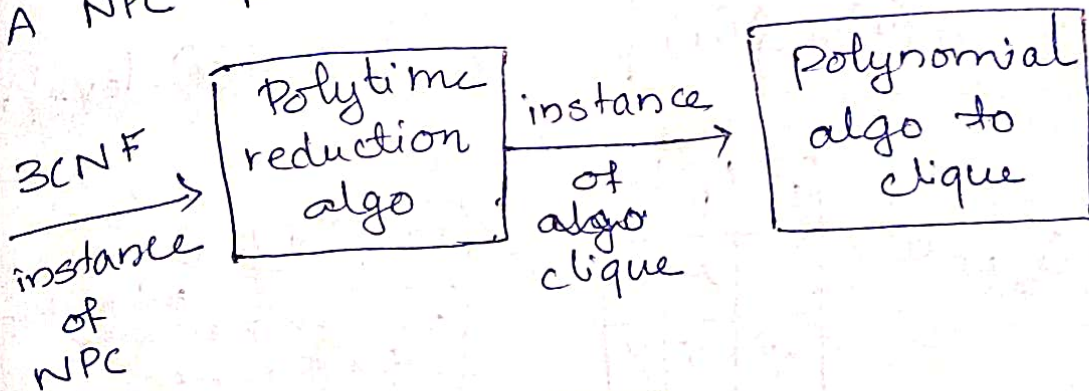
$$= 1 \wedge 1 \wedge 1 = \underline{\underline{1}}$$

Satisfiable. Hence proved that if we take CNF with k clause. Then we make a

clique of size k . Thus proof that clique problem is NP complete by the help of satisfiability problem.

CDP

1. Clique $\in$ NP
2. A NPC problem can be reduced to CDP



1/4

$n^2 - 1$ problem / 8 puzzle / 15 puzzle

1	2	3
	4	5
7	5	8



1	2	3
4	5	6
7	8	

Initial arrangement

goal arrangement

$$\hat{c}(x) = f(x) + \hat{g}(x)$$

$$f(x) = \text{level no}$$

Level 0, node 1

1	2	3
	4	6
7	5	8

$$c(1) = 0 + \underline{3} = 3$$

goal position
 3 → 1-1 ✓ 2-2 ✓ 3-3 ✓
 ① 4-5 × 6-6 ✓
 ② 7-7 ✓ 5-8 ×
 8 - Blank ③ ⇒ ③

$\hat{g}(x)$ - no of node non blank tile which are not in there goal position (check goal position)

Level 1

node 2

	2	3
1	4	6
7	5	8

$$c(2) = 1 + 4 = 5$$

node 3

1	2	3
4		6
7	5	8

$$c(3) = 1 + 2 = 3 \checkmark$$

node 4

1	2	3
7	4	6
	5	8

$$c(4) = 1 + 4 = 5$$

Level 2

node 5

1		3
4	2	6
7	5	8

$$c(5) = 2 + 3 = 5$$

node 6

1	2	3
4	6	
7	5	8

$$c(6) = 2 + 3 = 5$$

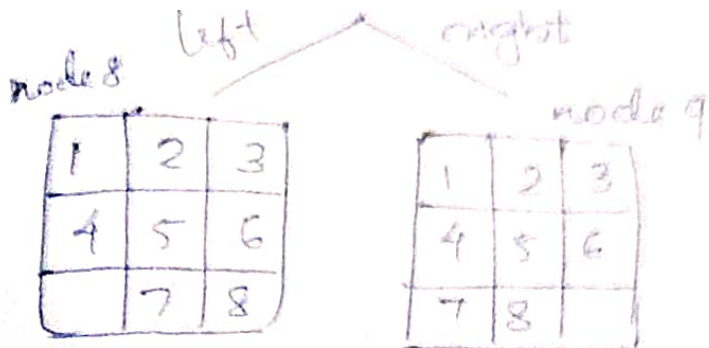
node 7

1	2	3
4	5	6
7		8

$$c(7) = 2 + 3 = 5 \text{ (min)}$$

no up is
 no de 7
 because it will
 become node 3

level 3

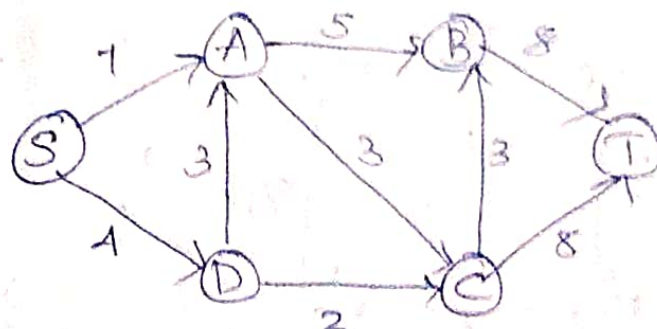


$$c(8) = 3 + 2 = 5$$

$$c(9) = 3 + 0 = 3$$

goal arrangement

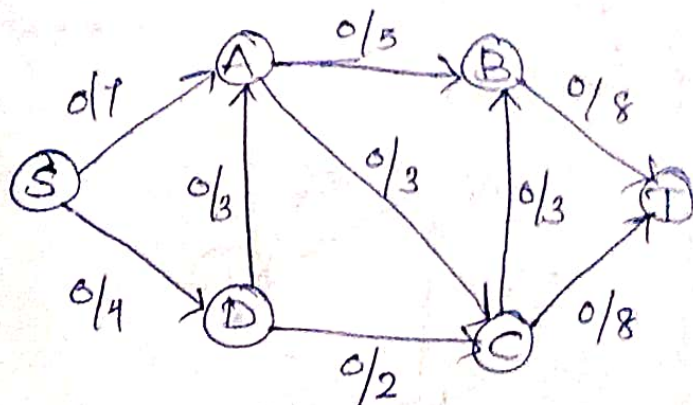
Ford Fulkerson Algorithm



S - source

T - Sink

Find augmented path

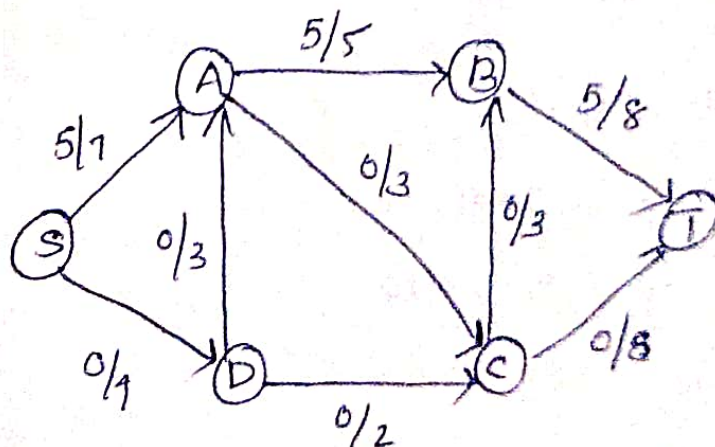


Path

S - A - B - T (min 0)

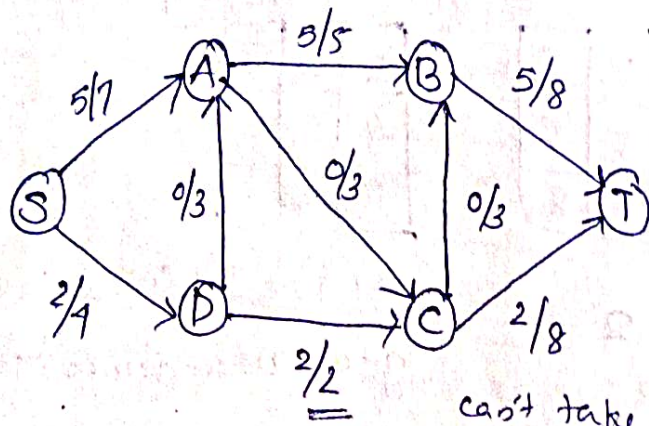
So, graph will be

path + 5

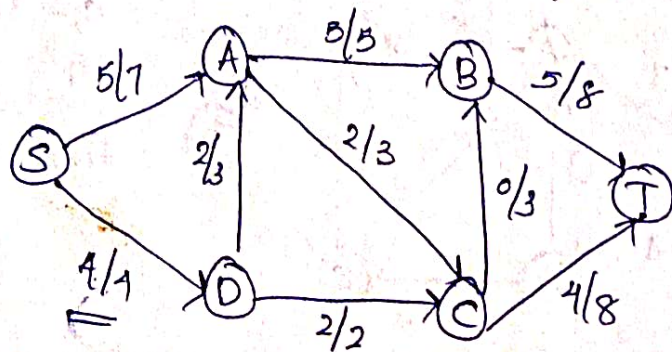


Can't take this path again because 5/5 full capacity.

Path S-D-C-T (min 2)

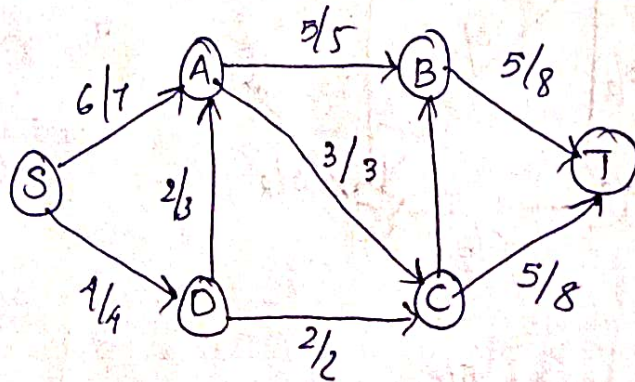


Path S-D-A-C-T (min 2)



$$\begin{aligned} 7-5 &= 2 \\ 3-2 &= 1 \rightarrow \text{min} \\ 8-4 &= 4 \end{aligned}$$

Path S-A-C-T (min 1)

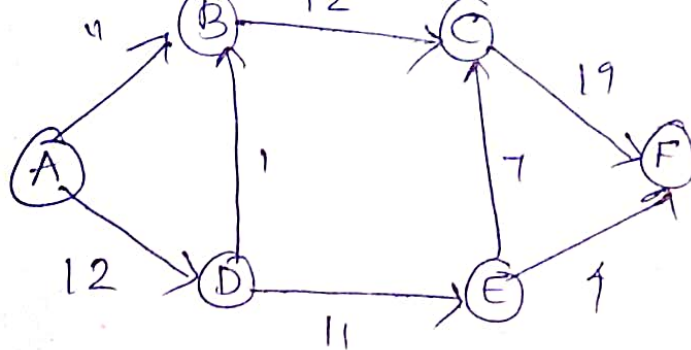


No path left to explore.

Total flow of S = 6 + 4 = 10

T = 5 + 5 = 10

HW



Total Ans.
Max flow of
A & F = 23

A - B - C - F (min 11)

A - D - E - F (min 4)

A - D - B - C - F (min 1)

A - D - E - C - F (min 7)